IT 340: BUSINESS INTELLIGENCE SYSTEMS

3.4 Project 3 – Writing SQL Queries

Submitted By:

Chinenye Edith Ugwu

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Lecturer:

Brother Mann

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Executive Summary

Analysis of the Most Used Plane Model at JFK International Airport

Introduction

This analysis was conducted to answer management's question: Which model of plane is the most used? The JFK flight dataset was analyzed using SQL to determine how frequently each aircraft model appeared in recorded flights departing from JFK International Airport.

For this analysis, the number of flights associated with each aircraft model was used as the primary measure of aircraft usage. The flight records were connected to the plane information using the aircraft tail number. The models were then grouped and counted to determine which model had the highest number of recorded flights.

SQL Query

```
SELECT
    plane.model,
    COUNT(flights.tailnum) AS plane_usage
FROM postgres.nycflights13.flights AS flights
JOIN postgres.nycflights13.planes AS plane
    ON flights.tailnum = plane.tailnum
WHERE flights.origin = 'JFK'
GROUP BY plane.model
ORDER BY plane_usage DESC
LIMIT 1;
```

Query Results (Top 10, for context)

Model	Number of Flights
A320-232	24,464
ERJ 190-100 IGW	16,816
CL-600-2D24	9,051
737-832	7,769
CL-600-2B19	6,199
767-223	4,257
A320-214	3,843
757-222	3,815
757-231	2,374
757-2Q8	2,280

Explanation of the Query

The query combines information from the flights table and the planes table using tailnum, which uniquely identifies an aircraft. It then limits the flight records to those departing from JFK International Airport.

The COUNT function calculates how many recorded flights were associated with each aircraft model. The GROUP BY statement organizes the records according to plane model, while ORDER BY arranges the

results from the most frequently used model to the least frequently used. Finally, LIMIT 1 returns only the aircraft model with the highest usage.

Running this query against the JFK flight data shows that the Airbus A320-232 is the most frequently used aircraft, with 24,464 recorded flights more than 7,600 flights ahead of the second-place ERJ 190-100 IGW. The result demonstrates how SQL can convert detailed flight records into information that management can use to understand aircraft utilization.

Why This Is a Good Metric

Using the number of recorded flights by aircraft model is a useful metric for several reasons.

First, the metric directly addresses the manager's question. If an aircraft model appears on more flights than other models, it provides a straightforward indication that the model is being used frequently in airport operations.

Second, the metric provides an objective and measurable comparison. Rather than relying on observation or assumptions, management can compare aircraft models using actual flight records and numerical results.

Third, understanding aircraft usage can support operational planning. Knowing which aircraft models operate most frequently may help management better understand demand for maintenance support, ground services, spare parts, and other resources associated with commonly used aircraft.

Limitations of the Metric

Although flight frequency is useful, it also has several limitations.

First, the metric measures the number of flights rather than the total amount of aircraft use. One aircraft model may complete many short flights, while another may complete fewer but much longer flights. Therefore, flight count alone does not measure total flight hours or distance travelled.

Second, the analysis does not consider how many individual aircraft of each model are available. A model may appear frequently simply because there are many aircraft of that model in operation. This does not necessarily mean that each individual aircraft is being used more intensively.

Third, the metric does not measure financial or operational performance. It does not show how much revenue a model generates, how many passengers it carries, its maintenance costs, fuel efficiency, delays, or reliability. Therefore, the most frequently used model is not automatically the most profitable or efficient model.

Additional Data That Would Improve the Metric

Additional information could provide management with a more complete understanding of aircraft utilization. Useful data would include the number of aircraft available for each model, total flight hours, distance flown, passenger numbers, seating capacity, revenue generated, fuel consumption, maintenance costs, delays, and cancellations.

For example, combining flight frequency with total flight hours would show not only how often a model flies but also how long it remains in operation. Passenger and revenue information could help determine whether highly used aircraft models are also contributing strongly to business performance.

Maintenance and delay information could further help management evaluate whether frequent usage creates additional operational costs or reliability concerns.

Conclusion

The analysis uses the number of recorded flights as a practical measure for determining which aircraft model is most frequently used at JFK International Airport. The SQL query connects flight records with aircraft information, counts usage by model, and identifies the model with the highest number of flights.

The results indicate that the Airbus A320-232 is the most frequently used aircraft model based on this measure, with 24,464 recorded flights. Flight frequency is a clear and useful starting point for understanding aircraft utilization, but management should consider additional measures such as flight hours, fleet size, passenger volume, revenue, maintenance costs, and delays before making broader operational decisions.